



Clustering-based Image-Text Graph Matching for Domain Generalization

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Abstract. Learning domain-invariant visual representations is important to train a model that can generalize well to unseen target task domains. Recent works demonstrate that text descriptions contain high-level class-discriminative information and such auxiliary semantic cues can be used as effective pivot embedding for domain generalization problems. However, they use pivot embedding in a global manner (i.e., aligning an image embedding with sentence-level text embedding), which does not fully utilize the semantic cues of given text description. In this work, we advocate for the use of local alignment between image regions and corresponding textual descriptions to get domain-invariant features. To this end, we first represent image and text inputs as graphs. We then cluster nodes within these graphs and match the graph-based image node features to the nodes of textual graphs. This matching process is conducted both globally and locally, tightly aligning visual and textual semantic sub-structures. We experiment with large-scale public datasets, such as CUB-DG and DomainBed, and our model achieves matched or better state-of-the-art performance on these datasets. The code is available at: <https://github.com/noparkee/Graph-Clustering-based-DG>

Keywords: Domain Generalization · Multimodal Learning

1 Introduction

How can humans effectively comprehend visual concepts despite variations in backgrounds, textures, and artistic styles? If it is impossible to collect sufficient examples of various combinations of domains, can current machine learning methods found on the i.i.d. assumption achieve robust generalization performance across domains? In this paper, we consider the domain generalization

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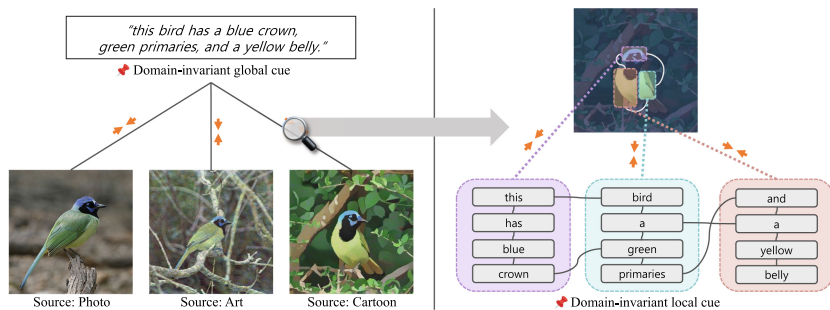


Fig. 1. Our model learns domain-invariant visual representations by matching images and text descriptions at both global and local levels. Images and texts are represented as clustering-based graphs, encouraging the model to learn domain-invariant local semantic cues (e.g., “a yellow belly” and “green primaries”).

problem on image datasets and introduce a novel clustering-based image-text graph matching to tackle the problem.

Domain generalization aims to improve a model’s generalization ability for unseen task domains. Previous research has explored various approaches to address this challenge, including minimizing domain discrepancies in the visual feature space [19, 35], augmenting data to cover various domains [18, 40], and utilizing ensemble learning [8, 22]. Notably, recent work such as GVRT [28] suggests leveraging natural language descriptions (e.g., “this bird has a blue crown, green primaries, and a yellow belly”) to infuse visual encoders with domain-invariant semantic cues, i.e., a visual encoder is optimized to produce an embedding that aligns well with the corresponding text embedding. While promising, optimizing a model with such global alignment often leads to suboptimal results, as these models may lack diverse attribute focus and occasionally attend to irrelevant regions for the class (e.g., see Figure 6a).

To address these limitations, as shown in Figure 1, we focus on local matching, wherein image regions are matched with corresponding textual descriptions (e.g., an image region of blue crown and “blue crown” in a sentence). This approach involves representing the text descriptions with graphs and aligning the embedding of images and text by matching the graphs.

As shown in Figure 2, the suggested method consists of three parts: (i) a graph-based visual encoder, (ii) a graph-based textual encoder, and (iii) a graph-based alignment. Based on the graph-based representations, we aim to learn the domain-invariant features by grounding the graph-based image features into textual graphs, as the textual graphs contain explicitly verbalized knowledge from humans’ typical reasoning. To solve the language grounding with structural information, we suggest a new method that clusters the graph node features then matches those clusters. By matching the multimodal graphs while clustering each

node’s features, our suggested method can get robust domain-invariant features representing multilevel semantic alignment.

Experimental results with two popular benchmark datasets, CUB-DG [28] and DomainBed [14], show the pivotal role of multimodal structural representations. Quantitatively, our suggested method achieves a new state-of-the-art performance, especially by increasing generalization ability on the most difficult domain *paint* of CUB-DG dataset. With robust qualitative visualization results, we argue that our model learns domain-invariant features across various feature resolutions by locally and globally aligning with textual graphs.

Our contributions can be summarized as follows. (1) We propose the first approach using graph representations for both image and text inputs for the DG problem. (2) We suggest a novel method that clusters and matches node features to align two multimodal graphs. (3) We achieve a new state-of-the-art DG performance on the CUB-DG dataset and DomainBed benchmark.

2 Related Work

Domain Generalization. Domain generalization aims to enhance a model’s ability to generalize to unseen target domains with different data distributions compared to the source domains. The main idea of domain generalization is to learn domain-invariant features from multiple source domains. Various methods have been proposed to resolve this problem (i) by reducing domain discrepancies in the feature space [19, 35], (ii) by implementing data augmentation [18, 40], and (iii) by utilizing ensemble learning [8]. (iv) Other studies have proposed using auxiliary semantic cues to facilitate learning domain-invariant features [2, 9, 33].

Recently, GVRT [28] successfully leverages textual descriptions for models to learn domain-invariant visual representations by aligning them with verbalized (domain-invariant and class-discriminative) knowledge from humans’ typical reasoning (e.g., given a text “this bird is black with an orange spot on its wing”). GVRT improves the model’s generalization power by leveraging visual and textual inputs together and simply matching global representations. However, our focus extends beyond this, emphasizing the alignment of locally-aware high-order semantic relations via graph structures.

Graph Neural Network. Along with the huge success of neural networks in computer vision and natural language processing domains, new methodologies to deal with irregular structural inputs have been recently suggested. Various graph-based neural network algorithms are recommended for learning representations from structural inputs like molecular graphs, social networks, and meshes. According to the ways of representing graph data, attention-based methods (e.g., MoNet [6]), convolution-based methods (e.g., GCN [20]), and message-passing methods (e.g., MPNN [13]) can be applied to graph representation learning. Those graph neural network methods have achieved great performance on graph-related tasks, such as node classification [4], link prediction [41], and graph classification problems [42], by leveraging the non-euclidean data manifolds to get

informative representations. Recently, the applications of graph neural networks have been extended to image and text domains [26]. By representing the image and text inputs as graphs, it becomes possible to consider the irregular and high-order correlations between tokens. In this paper, we suggest representing the multimodal inputs as graphs and matching the semantic correspondences between the multimodal inputs using graph neural networks to get the domain-invariant features.

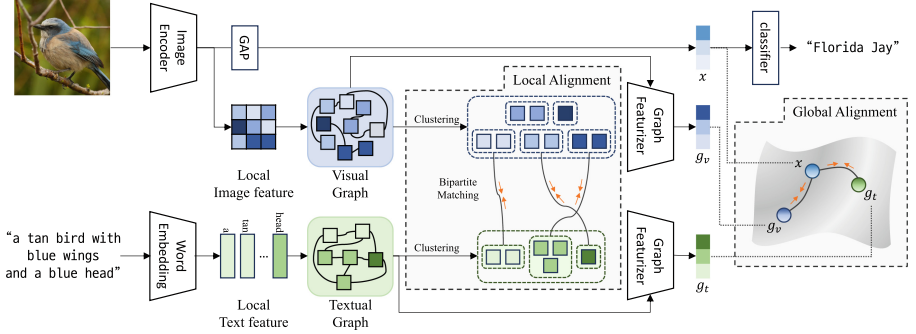


Fig. 2. An overview of our proposed method. We introduce multimodal graphs (visual and textual) that align with each other locally and globally, yielding domain-invariant visual features that are well-aligned with humans’ explicitly verbalized knowledge.

3 Method

Given a distribution over multiple (or single) source domains $\{\mathcal{S}_1, \mathcal{S}_2, \dots\} \in \mathcal{S}$, the domain generalization (DG) problem considers the following classical stochastic optimization, in which we minimize the data-dependent generalization upper bound of the expected task loss [34]:

$$\underset{\theta}{\text{minimize}} \quad \sup_{\mathcal{T}: D(\mathcal{S}, \mathcal{T}) \leq \rho} \mathbb{E}_{\mathcal{T}} [\mathcal{L}(\theta; \mathcal{S})] \quad (1)$$

where we consider unseen target domains $\mathcal{T} = \{\mathcal{T}_1, \mathcal{T}_2, \dots\}$ and the discrepancy between $\mathcal{S}$ and $\mathcal{T}$ is bounded by an arbitrary bound ρ , i.e. $D(\mathcal{S}, \mathcal{T}) \leq \rho$. We consider image classification scenarios and define the task-specific loss $\mathcal{L}$ function by the cross-entropy loss. Extracting domain-invariant representations from an image is key for training a model to generalize well to unseen domains.

Inspired by recent work by [28], we also want to improve the model’s generalization power by leveraging visual and textual inputs together. Our model learns to extract (domain-invariant) visual representations that are well-aligned with explicitly verbalized knowledge from humans’ typical reasoning. Unlike from [28] in that we focus more on aligning locally-aware high-order semantic relations via graphs instead of simply aligning global representations.

Figure 2 illustrates the architecture of our model, which is composed of three primary components: (i) a Graph-based Visual Encoder, (ii) a Graph-based Textual Encoder, and (iii) Graph-based Alignment for Learning Domain-Invariant Features. In (i), local latent representations (from a backbone network) are represented as a graph structure. Each local latent vector becomes a node, creating edges based on pairwise similarity in the embedding space (see Section 3.1). In (ii), we build a textual graph given a natural language description about a specific class (e.g., if the image corresponds to the *Florida Jay* class, the description could be “a tan bird with blue wings and a blue head.”). Each word embedding forms a node, creating edges based on embedding-level similarity (refer Section 3.2). In (iii), We regularize multi-modal encoders for locally aligned representations by minimizing graph-level distances between visual and textual data. Applying clustering-based graph matching makes our model generalizable by learning human-compatible visual cues. (see Section 3.3).

3.1 Graph-based Visual Encoder

Global Visual Feature Extraction. Following standards in the domain generalization task, we use the pretrained ResNet50 [16] on ImageNet dataset [11] as a backbone. Our backbone takes an image $\mathcal{I}$ as an input and produces a d -dimensional global visual representation $\mathbf{x}_g \in \mathbb{R}^d$. This global representation $\mathbf{x}_g$ is trained to predict its classification label $\mathbf{y}$ with a linear layer, yielding the per-class softmax probabilities $\hat{\mathbf{y}}$. Both the backbone and the classifier are trained by a classification loss $\mathcal{L}_c$ as follows:

$$\mathcal{L}_c(\mathbf{y}, \hat{\mathbf{y}}) = - \sum_i y_i \log(\hat{y}_i) \quad (2)$$

where $\mathbf{y} \in \mathbb{R}^{|C|}$ represents the ground-truth one-hot vector, and $|C|$ denotes the size of the ground-truth class set. The model minimizes the loss function $\mathcal{L}_c$, but, unfortunately, this optimization often results in the model becoming semantically shallow. Thus model would not generalize well in environments different from those in which they were trained. In our work, we aim to regularize our model to understand relations between visual cues and use those relations for the final verdict, thus making it more generalizable. We want to achieve such a regularization effect through utterances from human verbalized reasoning.

Locally-aware Visual Graph Construction. We first construct a graph with visual representations to achieve the above mentioned goal. Formally, given M number of d -dimensional local visual representations $\mathbf{x}_l \in \{\mathbf{x}_{l,1}, \mathbf{x}_{l,2}, \dots, \mathbf{x}_{l,M}\}$ extracted from intermediate layers of the backbone (before global average pooling layer), we consider these representations as a set of unordered nodes. Note that each representation vector $\mathbf{x}_{l,i} \in \mathbb{R}^d$ for $i \in [1, M]$ corresponds to a certain grid over an input image $\mathcal{I}$. Inspired by the recent work [15], we construct a graph such that each node $\mathbf{x}_{l,i}$ has an edge with the other K_v nearest neighbors (Visual Graph in Figure 2). We use the widely-used L_2 distance to measure

pairwise node similarity. In summary, our visual graph $\mathcal{G}_v = (\mathcal{V}_v, \mathcal{E}_v)$, where $\mathcal{V}_v$ denotes the node set consisting of M nodes (representing each visual feature of the local grid). The edge set $\mathcal{E}_v$ is comprised of the connections between nodes, with each node connected to its K_v nearest neighbors. Detailed explanations are provided in the supplementary material.

Graph-based Visual Representation. Given the visual graph $\mathcal{G}_v$, we further apply two layers of graph convolution network (GCN) [20], each followed by a linear layer, BatchNorm [17] layer and ReLU [1] activation. Subsequently, we employ dropout layer (only during training) and average readout operation to learn relational knowledge between local visual representations. Formally, we use a GCN-based function $f_{\text{GCN}}(\mathcal{G}_v)$ to obtain a final d_g -dimensional locally-aware visual graph representation $\mathbf{g}_v \in \mathbb{R}^{d_g}$, $\mathbf{g}_v = f_{\text{GCN}}(\mathcal{G}_v)$. Note that, we also add an additional classifier that takes the $\mathbf{g}_v$ as an input to create a graph that better captures the characteristics of the class. We provide detailed explanations in the supplementary material.

3.2 Graph-based Textual Encoder

Word-level Textual Graph Construction. Our graph-based visual representation $\mathbf{g}_v$ encodes relational knowledge via a graph structure between representations of local visual features $\mathbf{x}_l$. We empirically observe that such graph-based representation’s sole use is still insufficient for models to learn domain-invariant and human-compatible visual cues. Thus, to regularize our visual encoders to be aligned with human knowledge, we build a textual graph from a natural language description of each image, followed by aligning both visual and textual graphs. A sequence of L (at maximum) words is first tokenized and encoded with a standard word-level (learnable) embedding layer, producing d_t -dimensional embedding vectors $\mathbf{t} \in \{\mathbf{t}_1, \mathbf{t}_2, \dots, \mathbf{t}_L\}$ where $\mathbf{t}_i \in \mathbb{R}^{d_t}$. Similar to our Visual Graph $\mathcal{G}_v$, we consider these word embeddings as an unordered set. We then construct a graph such that each node $\mathbf{t}_i$ has an edge with the other K_t nearest neighbors (textual graph in Figure 2). We use L_2 distance to measure pairwise node similarity. Finally, we create a textual graph $\mathcal{G}_t = (\mathcal{V}_t, \mathcal{E}_t)$, where $\mathcal{V}_t$ denotes the node set comprising L nodes, each representing a textual feature of word embedding, and $\mathcal{E}_t$ signifies the edge set.

Graph-based Textual Representation. Given the textual graph $\mathcal{G}_t$, we apply the same architecture (but not shared) to obtain textual graph representation $\mathbf{g}_t \in \mathbb{R}^{d_g}$. I.e. we apply another GCN-based function $f_{\text{GCN}}(\mathcal{G}_t)$ to learn relational knowledge between word embeddings, $\mathbf{g}_t = f_{\text{GCN}}(\mathcal{G}_t)$.

3.3 Graph-based Alignment for Learning Domain-Invariant Features

We apply the following two graph-alignment approaches: (i) Graph-based Global Alignment and (ii) Local Alignment through Clustering-based Fine-grained Graph Matching, which comprises clustering and matching steps.

Global Graph Alignment. We assume that text descriptions inherently contain class-discriminative semantic cues. Thus, our model can learn domain-invariant features with aid of textual information. A standard approach to aligning different representations is minimizing the Euclidean distance. We employ this alignment technique to graph features as follows:

$$\mathcal{L}_{\text{global}} = \|f_{\text{proj},x}(\mathbf{x}_g) - f_{\text{proj},v}(\mathbf{g}_v)\|_2 + \|f_{\text{proj},x}(\mathbf{x}_g) - f_{\text{proj},t}(\mathbf{g}_t)\|_2 \quad (3)$$

where we use a linear layer to project each feature (i.e. $\mathbf{x}_g$, $\mathbf{g}_v$, and $\mathbf{g}_t$) such that these three projected features are pulled together. Note that $f_{\text{proj},x}$, $f_{\text{proj},v}$, and $f_{\text{proj},t}$ represent a projection layer. Importantly, as we use only a force to pull latent representations together, the training dynamics may become unstable, causing a representation collapse. To avoid this, like the approach in [28], we add an auxiliary classifier which is trained with the standard cross-entropy loss takes $f_{\text{proj},x}(\mathbf{x}_g)$ as an input to prevent a mode collapse, outputting the per-class softmax probability.

Clustering Graph Nodes. In addition to global alignment, initially, we tried to match the nodes in the visual graph with textual graph to align the locally-aware semantic relations. However, simply aligning nodes from two different graphs may not work as these nodes have different representations (i.e. a visual feature of a local image region vs. a word-level representation). Therefore, we present clustering-based local graph matching, which applies a node clustering algorithm to ensure that the two graphs have the same level of semantic representation and then performs graph matching. We define user-specified parameters $N_v (\leq M)$ and $N_t (\leq L)$ to the number of clusters for our visual and textual graphs, respectively. Note that we set $N_v \geq N_t$ since images may contain visual contents (e.g. backgrounds) that are not generally described in the text.

Our approach to constructing a graph is based on measuring node similarity, which can result in a well-defined semantic structure in the graph. Therefore, we choose a modularity-based method for graph clustering that can reflect this semantic structure while remaining stable. Specifically, we use a deep learning-based modularity measurement method [36]. Our model first encodes the cluster assignment matrix using the features of the graph nodes. Then, we calculate the modularity using this matrix, which measures the quality of the clustering. We train the model to maximize the modularity while also constraining it with collapse regularization to prevent trivial solutions such as assigning all nodes to

the same cluster. We formulate it as follows:

$$\mathcal{L}_d = -\frac{1}{2m} \text{Tr}(\mathbf{C}^T \mathbf{B} \mathbf{C}) + \frac{\sqrt{k}}{n} \left\| \sum_i \mathbf{C}_i^T \right\|_F - 1 \quad (4)$$

where $\mathbf{C}$ is the cluster assignment matrix calculated with our graph feature, and $\mathbf{B}$ is the modularity matrix calculated with the adjacency matrix. m , n , and k represent the number of edges, the number of nodes, and the number of clusters, respectively. The first term refers to modularity, which is measured using the assignment matrix and the modularity matrix, while the second term represents the collapse regularization term. So, our model can cluster semantically similar nodes together, allowing us to proceed with the matching process.

When applying node clustering to the visual graph $\mathcal{G}_v$, the cluster assignment matrix $\mathbf{C}$ has dimensions of $\mathbb{R}^{M \times N_v}$ (or $\mathbb{R}^{L \times N_t}$ when applied to the textual graph). Each element in matrix $\mathbf{C}$ at the i -th row and j -th column represents the softmax probability that the i -th local feature (or the i -th node) belongs to the j -th cluster. In this context, cluster feature is obtained through the following equation: $\mathcal{C}_v = f_{\text{proj},x}(\text{SeLU}((\mathbf{C}/N_v)^T \mathbf{x}_l))$, where SeLU [21] serves as one of the activation functions. Note that when dealing with the textual graph $\mathcal{G}_t$, N_v and $\mathbf{x}_l$ are replaced by N_t and $\mathbf{t}$, respectively.

Clustering-based Graph Matching. Inspired by previous work [7], we use the set-based loss, i.e. the bipartite matching loss, between two disjoint sets of clusters: (i) a set of clusters $\mathcal{C}_v \in \{\mathcal{C}_v^1, \mathcal{C}_v^2, \dots, \mathcal{C}_v^{N_v}\}$ of the visual graph $\mathcal{G}_v$ and (ii) a set of clusters $\mathcal{C}_t \in \{\mathcal{C}_t^1, \mathcal{C}_t^2, \dots, \mathcal{C}_t^{N_t}\}$ from the textual graph $\mathcal{G}_t$. We minimize the following pair-wise matching loss:

$$\mathcal{L}_p = \frac{1}{N_t} \sum_{i=1}^{N_t} \|\mathcal{C}_v^{\mu_i} - \mathcal{C}_t^i\|_2 \quad (5)$$

where $\mu_i \in \{1, 2, \dots, N_v\}$ is the cluster index of $\mathcal{C}_v$ which matches to i in $\mathcal{C}_t$, producing the smallest total Euclidean distance by bipartite matching.

As the pair-wise matching loss pulls positive pairs together, negative pairs to add a repulsive force may need to prevent representation collapse. Thus, we also use a hinge loss based on $\mathcal{C}_v^i$ and $\mathcal{C}_t^j$ (where $i \in [1, N_v]$ and $j \in [1, N_t]$), considering them as a negative pair if they are clusters for different input images. Thus, the matched distance $\mathcal{L}_p$ should be smaller than any other pairs between $\mathcal{C}_v^j$ and $\mathcal{C}_t^i$ (or $\mathcal{C}_v^j$ and $\mathcal{C}_t^i$). We formulate it as a hinge loss as follows:

$$\mathcal{L}_h = \max(0, \mathcal{L}_p - \text{MinDst}(\mathcal{C}_v', \mathcal{C}_t) + \epsilon) + \max(0, \mathcal{L}_p - \text{MinDst}(\mathcal{C}_v, \mathcal{C}_t') + \epsilon) \quad (6)$$

where $\text{MinDst}(\mathcal{C}_v, \mathcal{C}_t')$ and $\text{MinDst}(\mathcal{C}_v', \mathcal{C}_t)$ represents the minimum pair-wise matching loss similar to $\mathcal{L}_p$, but is applied between two disjoint sets of clusters originating from different inputs within a mini-batch. We compute it across all sample pairs in a mini-batch and use the average as the final loss value:

$$\mathcal{L}_{\text{local}} = \frac{1}{B} \sum_b (\lambda_d \mathcal{L}_d + \lambda_h \mathcal{L}_h + \lambda_{\text{aux}} \mathcal{L}_{\text{aux}}) \quad (7)$$

where we set the size of a mini-batch to B and λ_p , λ_h , and λ_d adjustable hyper-parameters that control the weight of each loss term. In our model, values of 1, 0.1, and 0.1 are used for λ_d , λ_h , and λ_{aux} , respectively. Note that, similar to our global alignment module, we also add an auxiliary classifier that takes the average-pooled representation of visual clusters matched with textual clusters, denoted as $\frac{1}{N_t} \sum_{i=1}^{N_t} \mathcal{C}_v^{\mu_i}$, as an input. This classifier outputs the per-class softmax probability and is trained using the standard cross-entropy loss $\mathcal{L}_{\text{aux}}$.

Loss Function. Ultimately, we train our model end-to-end by minimizing the following loss L :

$$\mathcal{L} = \mathcal{L}_c + \mathcal{L}_{\text{global}} + \mathcal{L}_{\text{local}} \quad (8)$$

4 Experiments

4.1 Setup

Implementation Details. Same as previous domain generalization approaches, we also use ResNet50 [16], pre-trained on ImageNet [11], as our backbone, yielding a 2,048-dimensional visual representation from the last layer. Our model is trained end-to-end for 5,000 training steps using Adam optimizer with a learning rate of 5e-5. For training, we use standard image augmentation techniques such as random cropping, horizontal flipping, color jittering, grayscale conversion, and normalization. Our implementation is based on DomainBed [14], which is a unified domain generalization testbed, and our code will be publicly available upon publication. More details, including information that varies depending on the dataset, are available in supplementary material.

Datasets. To demonstrate our model’s effectiveness, we first use the CUB-DG dataset (for fine-grained image classification task), which is extended from the CUB dataset [39] for the domain generalization task. This dataset contains 11,768 images for 200 classes of North American bird species. Each image has 10 text descriptions describing the content in detail, e.g., “this bird is black with an orange spot on its wing.” Each image is manipulated to create the following four domains: Photo, Cartoon, Art, and Painting. We follow the common evaluation protocol and use the CUB-DG dataset’s official split (the train and validation set has 5,994 samples, while the test set has 5,794 samples).

Further, we also evaluate our model on DomainBed [14], which contains the following five multi-domain DG datasets: VLCS [12], PACS [25], OfficeHome [38], TerraIncognita [3], and DomainNet [23]. Among these, we would emphasize that PACS [25] dataset is useful for our experiments as (i) it provides a bigger domain shift than existing photo-only benchmarks, and (ii) it needs to exploit local

information to learn discriminative subtle visual features. We follow the standard evaluation protocol used in [14]. For datasets that do not provide text inputs, we use both (1) textual class definitions from the Oxford dictionary similar to GVRT [28] and (2) descriptions generated by InstructBLIP [10] with the prompt “write a detailed description about the image.” (refer supplementary material).

4.2 Performance Comparison

Table 1. The out-of-distribution classification accuracies (in %) on CUB-DG (top) and PACS (bottom) datasets based on the standard leave-one-out multi-source DG task setting. We compare ours with other existing DG approaches. (we provide full tables in supplementary material). *Abbr.* I: Image, T: Text.

Algorithms	Modality	Target Domain (Data: CUB-DG [28])				Avg. $\uparrow$
		Photo	Cartoon	Art	Paint	
MIRO [9]	I	68.2	59.1	46.5	38.2	53.0
SD [31]	I	71.3	62.2	50.8	34.8	54.7
CORAL [35]	I	72.2	63.5	50.3	35.8	55.4
CCFP [24]	I	70.0	61.5	52.1	<u>40.4</u>	56.0
GVRT [28]	I+T	<u>74.6</u>	<u>64.2</u>	<u>52.2</u>	37.0	57.0
Ours	I+T	75.4	65.5	54.0	41.4	59.1
Algorithms	Modality	Target Domain (Data: PACS [25])				Avg. $\uparrow$
		Art Painting	Cartoon	Photo	Sketch	
SelfReg [19]	I	<u>87.9 $\pm$ 1.0</u>	79.4 $\pm$ 1.4	96.8 $\pm$ 0.7	78.3 $\pm$ 1.2	85.6
CORAL [35]	I	88.3 $\pm$ 0.2	80.0 $\pm$ 0.5	97.5 $\pm$ 0.3	78.8 $\pm$ 1.3	86.2
mDSDI [5]	I	87.7 $\pm$ 0.4	80.4 $\pm$ 0.7	98.1 $\pm$ 0.3	78.4 $\pm$ 1.2	86.2
SagNet [29]	I	87.4 $\pm$ 1.0	80.7 $\pm$ 0.6	97.1 $\pm$ 0.1	80.0 $\pm$ 0.4	86.3
CCFP [24]	I	87.5 $\pm$ 0.1	<u>81.3 $\pm$ 0.3</u>	96.4 $\pm$ 0.3	81.4 $\pm$ 0.8	86.6
Ours	I+T	<u>87.9 $\pm$ 0.7</u>	81.4 $\pm$ 0.1	<u>98.0 $\pm$ 0.1</u>	<u>80.5 $\pm$ 1.1</u>	87.0

As shown in Table 1, we compared the out-of-distribution classification accuracies on the following two datasets: CUB-DG (top) and PACS (bottom) datasets. We compare ours with other existing state-of-the-art domain generalization approaches, SagNet [29], MIRO [9], SD [31], CORAL [35], GVRT [28], SelfReg [19], mDSDI [5], and CCFP [24]. Due to space constraints, we only report top-6 results (we provide full tables in supplementary material).

As shown in Table 1 (top), our proposed method clearly outperforms the other domain generalization techniques on the CUB-DG dataset in all target domains with a significant gain. In terms of the average image classification

accuracy, ours shows 59.1%, which is 2.1% higher than GVRT [28] (which uses the same image and text inputs) and 3.1% higher than image-only approach, CCFP [24]. Similar trends are also observed in our experiment on the large-scale PACS [25] dataset. As shown in Table 1 (bottom), our model also outperforms the other approaches, i.e., ours shows 87.0% that is 0.4% higher than CCFP [24]-based SOTA approach and 1.9% higher than GVRT [28]. These confirm that our graph-based approach is effective in aligning visual and textual encoders for fine-grained image classification tasks, improving the visual encoder’s generalization power to unseen target domains.

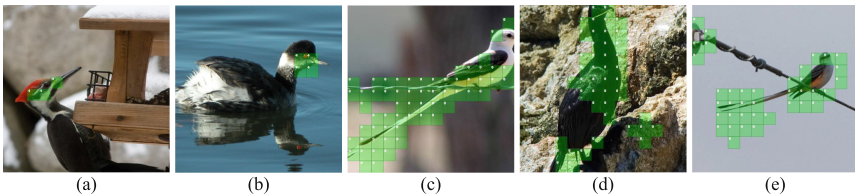
4.3 Few-shot DG Performance Comparison

Conventional DG approaches often assume that a sufficient number of images is available for all classes and domains enough to learn domain-invariant class-discriminative features. However, this may be practically challenging in

real-world scenarios. We emphasize that our method, which leverages textual descriptions as pivotal information, can benefit learning domain-invariant features in the few-shot setting. In Table 2, our model significantly outperforms the other approaches on PACS [25] and VLCS [12] datasets. Note that we use randomly chosen five images (per class in each domain) as an input to train all models (i.e., 5-shot DG). Except for the number of training images, we generally follow the standard protocol of DomainBed for evaluation.

Table 2. Few-shot DG performance comparison.

Algorithm (Data: PACS [25])	Avg.	Algorithm (Data: VLCS [12])	Avg.
mDSDI [5]	63.5	mDSDI [5]	68.5
CORAL [35]	64.6	GVRT [28]	69.4
MIRO [9]	65.5	MIRO [9]	69.6
GVRT [28]	68.7	CORAL [35]	71.1
Ours	70.7	Ours	71.4



- (a) This medium sized bird **has a red crown**, black superciliary and white check patch.
- (b) This bird **has** wings that are black **and has orange eyes**.
- (c) This bird **has a** white head **and chest** that slowly turns in to a peach color near his feet, and has extremely long tail feathers that are longer than his body.
- (d) This bird **has** wings that are black **and has a long neck**.
- (e) This bird **has** wings that are brown with very long tailfeathers.

Fig. 3. Examples of the matched image region (in visual graph clusters) and texts (in textual graph clusters).

4.4 Analysis on CUB-DG Dataset

Analysis of Graph Clusters and Their Matchings. In Figure 3, we provide examples of a matched pair of image regions and a set of words. For example, in (a), a region around the bird’s head is matched with a textual graph cluster that contains words including “orange eyes”. We observe that our model reasonably matches image features with class-discriminative texts, e.g., red crown, cheek patch, extremely long tail, and long neck.

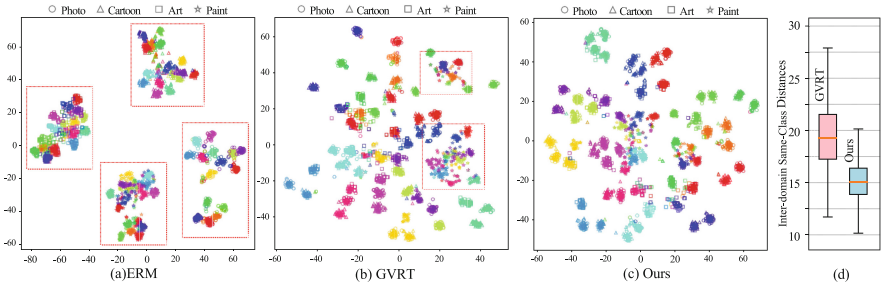


Fig. 4. Visualizations by t-SNE for (a)ERM [37], (b)GVRT [28], and (c)Ours on CUB-DG. Points are color-coded differently by its class and has different shapes according to its domain. (d)We also compare inter-domain same-class distances.

As shown in Figure 4, we provide t-SNE [27] visualization of (a) ERM [37], (b) GVRT [28], and (c) Ours to visualize their embedding space on CUB-DG dataset. We use different marker styles (for target domains) and different colors (for classes). An ideal model would show that visual features of the same class but different domains are gathered together. Ours clearly outperform ERM, which has scattered points per domain and better than GVRT in that features of the same class but different domains are more clustered (see red boxes). In Figure 4 (d), we provide box plots for GVRT and ours, showing that our model produces lower same-class inter-domain distances than GVRT. Note that we provide detailed t-SNE visualizations in supplementary material. Figure 5 further shows the nearest examples in the unseen target domain to the given image (e.g., “dog”). In contrast to MIRO and GVRT, which often provide examples of different classes (e.g., “person”), ours consistently provide examples of the same class. This is consistent with our t-SNE analysis. We provide more examples in in supplementary material.



Fig. 5. Exemplars of the nearest examples from PACS dataset (in the unseen target domain) to the given image (e.g., “dog” and “horse”).

GradCAM Visualization. In Figure 6, we use GradCAM [32] to visualize image regions where the model focuses on for the final verdict. We observe that our model generally focuses on multiple class-discriminative features, giving the benefits of more robust and generalizable recognition performance. More specifically, in the first image of Figure 6b, our model focuses on the head region, a relevant area for the classification task, in contrast to the GVRT model which attends to a region less relevant to class information (Figure 6a).

Furthermore, in the second image of the GVRT scenario, the model concentrates solely on the belly region, while our model exhibits a more diverse focus on attributes, encompassing both the belly and beak. Unlike GVRT, which focuses solely on global alignment, our model incorporates local alignment through graphs. This approach enables our model to capture diverse attributes, increasing its ability to generalize effectively.



Fig. 6. GradCAM [32] visualizations to evaluate where the model sees.

Table 3. Ablation studies to compare variants of our model. Data: CUB-DG.

Global Alignment	Local Alignment	Visual Graph	Textual Graph	Target Domain				Avg.
				Photo	Cartoon	Art	Paint	
-	-	✓	✓	65.1	52.5	38.2	29.0	46.2
✓	-	-	-	69.5	57.1	44.2	30.2	50.2
✓	-	✓	-	70.3	57.0	48.1	33.5	52.2
✓	-	-	✓	75.0	64.4	53.0	34.7	56.8
-	✓	✓	✓	71.4	57.6	46.6	37.2	53.2
✓	✓	✓	✓	75.4	65.5	54.0	41.4	59.1

Ablation Studies. In Table 3, we conduct an ablation study to demonstrate the effect of main modules: (i) a global alignment, (ii) a local alignment, (iii) a visual graph, and (iv) a textual graph. Our study demonstrates that (1) a global alignment, which aligns graph-level features together, effectively improves accuracies, especially in photo, cartoon, and art domains. (2) Adding local alignment, which aligns graphs via the clustering-based matching algorithm, improves all domains while using both alignments outperforms the alternatives. (3) Either using a visual or textual graph alone improves model generalization, but the gain is marginal with the visual graph alone. (4) The gain is maximized by using both graphs, which indicates that a graph structure effectively transfers text knowledge to train a generalizable visual encoder.

Table 4. The test accuracies (in %) on the DomainBed datasets in the multi-source DG task setting. We compare ours with other existing DG approaches (we provide full tables in supplementary material). †: utilizing texts from a dictionary, ‡: incorporating texts from InstructBLIP. *Abbr.* I: Image, T: Text.

Algorithm	Modality	Dataset					Avg.
		VLCS	PACS	OfficeHome	TerraIncognita	DomainNet	
CORAL [35]	I	78.8 ± 0.6	86.2 ± 0.3	68.7 ± 0.3	47.6 ± 1.0	41.5 ± 0.1	64.6
CCFP [24]	I	78.9 ± 0.3	86.6 ± 0.2	68.9 ± 0.1	48.6 ± 0.4	41.2 ± 0.0	64.8
mDSID [5]	I	79.0 ± 0.3	86.2 ± 0.2	69.2 ± 0.4	48.1 ± 1.4	42.8 ± 0.1	65.1
GVRT (PTE) [28]	I+T	79.0 ± 0.2	85.1 ± 0.3	70.1 ± 0.1	48.0 ± 0.2	44.1 ± 0.1	65.2
MIRO [9]	I	79.0 ± 0.0	85.4 ± 0.4	70.5 ± 0.4	50.4 ± 1.1	44.3 ± 0.2	65.9
Ours [†]	I+T	78.3 ± 0.4	85.7 ± 0.1	70.1 ± 0.1	49.5 ± 0.9	43.7 ± 0.0	65.5
Ours [‡]	I+T	78.6 ± 0.3	87.0 ± 0.4	70.4 ± 0.2	49.2 ± 0.5	44.2 ± 0.0	65.9

Table 5. Performance comparison between variants of our model with different matching techniques: bipartite matching and greedy matching. Data: CUB-DG.

Bipartite Matching	Greedy Matching	Time Complexity	Target Domain				Avg.
			Photo	Cartoon	Art	Paint	
-	✓	$O(V^2)$	75.3	64.7	54.8	36.6	57.8
✓	-	$O(V^3)$	75.4	65.5	54.0	41.4	59.1

Complexity of Graph Matching. In our matching algorithm design, we explored two approaches: bipartite matching and greedy matching. Bipartite matching establishes one-to-one correspondences between clusters, minimizing pairwise distances, while greedy matching allows many-to-one associations based on spatial proximity. Bipartite matching operates with a time complexity of $O(V^3)$, where V denotes the number of vertices, while greedy matching operates in $O(V^2)$. Despite the higher time complexity of bipartite matching, our experimental results (refer to Table 5) demonstrate its superior performance over greedy matching. This may be attributed to the constraints imposed on one-to-one matching, which result in a dispersed effect on models attempting to optimize cluster pairs on a global scale. Moreover, given that V does not exceed 5 in our method, we opted for bipartite matching due to its enhanced performance in our specific context.

4.5 Performance on DomainBed Benchmark

We evaluate our model with a large-scale DomainBed [14] datasets. We use the following five multi-domain datasets, including VLCS [12], PACS [25], OfficeHome [38], TerraIncognita [3], and DomainNet [30], comparing ours with 19 domain generalization algorithms. Due to space constraints, we only report top-6 results (see supplementary material for full table). The reported score rep-

resents averaged results obtained from three independent runs using randomly chosen hyperparameters. We observe in Table 4 that our proposed method shows matched or better state-of-the-art performance, where it ranks 1st (tied) in average performance.

5 Conclusion

We propose a novel domain generalization method that encodes domain-invariant visual representations. To this end, we use a textual description to utilize verbalized (domain-invariant) knowledge from humans' typical reasoning. To align these, we use a clustering-based graph-matching algorithm based on visual and textual graphs built upon images and texts, respectively. We evaluate our model with state-of-the-art domain generalization approaches on CUB-DG and DomainBed datasets, achieving SOTA performance.

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